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AA216 : Flight Mechanics and Classical Control

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Problem Statement: Sloshing suppression in liquid transport systems using feedback control

The project aims to design and implement an effective control system to suppress liquid sloshing in a moving container. When a container is subjected to motion (acceleration), the liquid inside exhibits oscillatory behavior, leading to instability, spillage, and reduced system performance.

The system is modeled in a relative frame of reference, where liquid motion is analyzed with respect to the moving container and approximated as a pendulum system. The objective is to control the container motion (input acceleration/position) such that the sloshing amplitude (liquid displacement) is minimized.

To achieve this, a closed-loop control system using a PID controller is implemented, where displacement error is used as feedback to regulate motor input. The controller modifies the motion profile to reduce oscillations, improve damping, and ensure stable and efficient transport of liquid.

Control Objective

The **overall objective** is to design a feedback controller for the linearized sloshing model that stabilizes the system, suppresses oscillations, and improves transient and steady-state performance under external disturbances.

- **Towards Stability:** From a stability standpoint, the system focuses on reducing liquid sloshing and maintaining stable motion during transport. A closed-loop control system using a PID controller is implemented to minimize oscillations. It ensures smooth operation even under disturbances and varying conditions. Stability is achieved by controlling the dynamic behavior of the liquid.
- **Towards Performance:** From a performance standpoint, the objective is to minimize the amplitude of fluid oscillations, reduce settling time, and improve damping characteristics of the system. The PID controller provides faster response, reduced overshoot, and improved accuracy compared to uncontrolled systems.
- **Objective explanation:** The project involves modelling the liquid system as a pendulum and designing a PID controller. A closed-loop system is implemented for better control. The control goal is to regulate slosh displacement towards zero and achieve short settling time with adequate damping, thereby stabilizing the system and minimizing oscillations. The goal is to enhance safety and efficiency in industrial fluid transport.

System block diagram

- Non-linear governing equation

$$J \frac{d^2\theta}{dt^2} = -c \frac{d(l\theta)}{dt} l \cdot \cos^2 \theta - mgl \sin \theta + m\alpha l \cos \theta$$

- Linearised equation

$$\ddot{\theta} = -\frac{c}{m}\dot{\theta} - \frac{g}{l}\theta + \frac{1}{l}\alpha$$

- natural frequency and damping ratio

$$\omega_n = \sqrt{\frac{g}{l}} \quad \zeta = \frac{c}{2m} \sqrt{\frac{l}{g}}$$

- Transfer function and poles

$$\frac{h(s)}{\alpha(s)} = \frac{\frac{l}{l}}{s^2 + \frac{c}{m}s + \frac{g}{l}}$$

$$s = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{g}{l}}$$

Project Report

- **Plant Transfer Function:**

The sloshing system is modeled using a linearized pendulum approximation in the relative frame of reference. The transfer function relating liquid height to container acceleration is given by:

$$h(s)/\alpha(s) = \frac{0.5s^2 + 4.722s + 1635}{s^2 + 4.722s + 16350.5}$$

The system has complex conjugate poles, indicating an oscillatory but stable behavior.

- **Important Figure: (Step Response of Liquid Sloshing System with PID Control)**



The figure shows the step response of the liquid sloshing system, where the output exhibits oscillatory behavior initially due to low damping. The amplitude of oscillations gradually decreases with time, indicating a stable but underdamped system. Eventually, the response settles to a steady value, demonstrating effective damping and sloshing suppression.

PID Controller Based System Stability Analysis

A PID controller is used to improve the stability and performance of the system. The proportional action helps in increasing the response speed, the integral action eliminates steady-state error, and the derivative action reduces oscillations, which is especially important in sloshing control.

The system is inherently stable since its poles lie in the left half of the complex plane. However, it is underdamped and exhibits large oscillations in its response. To improve stability and overall performance, a closed-loop PID controller is implemented. This increases the damping of the system and shifts the poles further into the left half plane, resulting in a more stable response.

The system performance is evaluated based on key parameters such as rise time (t_r), settling time (t_s), maximum overshoot (M_p), and steady-state error. These parameters help in analyzing how effectively the system responds over time.

After applying the PID controller, the system shows improved behavior with reduced oscillations and decreased settling time. The controller modifies the system dynamics by increasing damping and shifting the poles further into the left half of the complex plane. This results in a faster and more stable response.

The analysis and design of the system are carried out using the following tools:

- BIBO Stability (to ensure output remains bounded for bounded input)
- Root Locus Method (to study pole movement with controller gain)
- Routh Stability Criterion (to verify system stability)
- Steady-State Error Analysis (to evaluate system accuracy)
- MATLAB/Simulink (for simulation and analysis)

Conclusion:

The PID controller effectively improves the system performance by reducing oscillations and enhancing stability, ensuring a faster and controlled response.

List of components you would use in this experiment

Actuators :

- **PMDC Motor:** Drives the belt and provides motion to the container
- **Gearbox:** Reduces speed and increases torque for smooth operation.

Sensors:

- **Encoder (built-in with motor):** Measures speed and position of container
- **High-Speed Camera:** Captures liquid surface (sloshing behavior)
- **Limit Switches:** Detect end position and ensure safety

Mechanical Components:

- **Liquid-filled Container:** Holds the liquid where sloshing behavior is observed.
- **Linear Guide Rail (HIWIN LGR):** Provide smooth, low-friction linear motion for the system.
- **Carriage (moving platform):** Supports and carries the container along the path.
- **Base Frame (wood/structure):** Provides structural support and stability to the setup.
- **Belt drive system (belt + pulleys):** Transfers motion from motor to carriage.

Control & Electrical:

- **Controller (PID via MATLAB/Simulink):** Controls motion to reduce sloshing and improve stability.
- **Power Supply:** Provides electrical energy to all components
- **Motor driver circuit:** Regulates voltage/current to control motor speed.

EXTRA INFORMATION

Fluid sloshing control has significant importance in various real-world engineering systems where liquid motion can affect stability and performance. Some of the real-world applications are as follows:

- In **tank trucks and fuel transport vehicles**, where sloshing can lead to rollover accidents due to shifting liquid mass and dynamic forces. Advanced modeling using pendulum-based systems helps in predicting and controlling such behavior, improving vehicle safety.
- In **aerospace systems**, particularly in spacecraft carrying liquid fuel. Sloshing of propellant can disturb the attitude and orientation of the spacecraft, affecting navigation and control. Research shows that modeling the liquid as an equivalent pendulum system allows effective controller design for stabilizing spacecraft motion under disturbances.
- In **industrial liquid handling systems and robotic transport**, where containers are moved along predefined paths. Sloshing can lead to spillage, inaccurate positioning, and reduced efficiency. Controlling sloshing ensures smoother operation and improved precision in automated systems.

Limitations and Improvement:

- **Simplified Pendulum Model:**
The equivalent pendulum model does not fully capture complex fluid dynamics.
Improvement: Use multi-mode or higher-order models for better accuracy.
- **Linearization Assumption:**
The model is valid only for small-angle oscillations and fails under large disturbances.
Improvement: Apply nonlinear control techniques such as adaptive or sliding mode control.
- **Parameter Sensitivity:**
System performance varies with changes in fluid level, tank shape, and disturbances.
Improvement: Design robust controllers and use state estimation methods like observers.
- **Limited Adaptability:**
Conventional controllers may not perform well under varying real-time conditions.
Improvement: Incorporate intelligent control methods such as fuzzy logic or machine learning.

Conclusion:

In this project, fluid sloshing in a partially filled container is modeled using an equivalent pendulum approach and controlled using a feedback control system. The study highlights the importance of controlling sloshing to ensure system stability and improve dynamic performance under external disturbances. While the simplified linear model provides a good basis for control design, real-world applications may require more advanced modeling and control strategies to handle nonlinearities and uncertainties. Overall, the proposed approach demonstrates an effective method for reducing sloshing oscillations and enhancing system stability, with potential applications across transportation, aerospace, and industrial systems.